



PrajapatiSamaj — Community Matrimonial Platform

Full-Stack Enterprise Web Portal with 3-Tier Monetization & Traditional Marriage Biodata Studio

An end-to-end, high-security matrimonial solution purpose-built for the Prajapati community. Designed with a strict layered architecture in Java 17 & Spring Boot 3 on the backend, complemented by a lightning-fast React + Vite frontend. Features frictionless OTP registration, 60 FPS HTML5 canvas photo cropping, multi-criteria discovery, mutual interest matchmaking, Razorpay monetization, and authentic 1-page A4 Marriage Biodata generation.

Developer: Pragnesh Maru (Java Spring Boot Developer)

Architecture: Layered Controller-Service-Repository + DTO Mapper

Backend: Spring Boot 3.2.5 • Java 17 • MySQL

Frontend: React 18 • Vite • Modern Dark Mode UI

80+

PRODUCTION FEATURES

8 Themes

A4 BIODATA STUDIO

3 Tiers

MONETIZATION SYSTEM

< 50ms

HIKARICP QUERY POOL

INDUSTRY CHALLENGES ADDRESSED

- Privacy Violations:** Public platforms expose phone numbers and home addresses to scammers without owner consent.
- Inaccurate Matchmaking:** Generic filters miss Gujarati community nuances (Gotra, Mangal/Shani status, Kundali birth time).
- Manual Biodata Hassle:** Families spend hours formatting Word documents that look amateurish or span multiple clumsy pages.
- Fake & Abandoned Profiles:** Free platforms become dumping grounds for unverified and inactive registrations.

ENGINEERED ARCHITECTURAL SOLUTIONS

- Masked Privacy & Tier-Gated Reveal:** Phone numbers are masked ("98*****") by default; Level 3 VIP tier unlocks verified access.
- Culturally Tailored Schemas:** Native support for Gotra, 3-state Mangal/Shani toggles, 2-decimal weight, and birth time selectors.
- Protected Biodata Studio:** Auto-fills 8 luxury traditional themes with anti-screenshot/anti-recording safeguards.
- Two-Step OTP & Freshness Tracking:** Active time tracking ("Active today", "Active 3d ago") with mandatory email OTP verification.

FULL-STACK TECHNOLOGY STACK & ECOSYSTEM

Layer	Technologies & Libraries	Key Responsibility / Architecture Value
Backend Core	Java 17, Spring Boot 3.2.5, Spring MVC, Spring Data JPA	High-performance REST API services, transaction management (<code>@Transactional</code>), and async tasks (<code>@EnableAsync</code>).
Security & Auth	Spring Security 6, JWT (0.11.5), BCrypt Password Encoder	Stateless JWT Bearer token authentication (24h expiry), role enforcement, and secure salted hashing (strength 12).
Database & Pool	MySQL 8.0, HikariCP Enterprise Pool, Hibernate 6 ORM	Optimized connection pooling (25 max connections, leak detection), automated schema initialization, and transactional consistency.
Frontend Core	React 18, Vite Bundler, React Router v6, Axios	Single Page Application (SPA), ultra-fast HMR builds, permanent dark-mode theme, dynamic host binding for local Wi-Fi mobile testing.
Imaging & Studio	HTML5 Canvas API, requestAnimationFrame 60FPS Sync	Zero-lag client-side photo cropping, zoom, 90° rotation, 4:5/1:1/16:9 aspect ratios, and anti-screenshot watermark overlays.
Integrations	Razorpay Payment Gateway, JavaMailSender SMTP	Secure payment order generation & HMAC-SHA256 signature verification, transactional HTML OTP emails, and registration notifications.



STRICT LAYERED ARCHITECTURE & SEPARATION OF CONCERNS

The backend strictly separates concerns across independent architectural tiers, ensuring zero business logic contamination inside controllers and avoiding entity leaks to client consumers.

1. WEB / API

Controller Layer

Validates client payloads (`@Valid`), extracts auth principals, handles HTTP status codes, and delegates to services.

2. BUSINESS CORE

Service Layer

Encapsulates business rules, transactional boundaries, quota calculations, email notifications, and mutual match state transitions.

3. DATA ACCESS

Repository Layer

Spring Data JPA repositories, JPQL custom joins, CriteriaBuilder dynamic specifications (`ProfileSpecification`).

4. TRANSFORMATION

DTO & Mapper Layer

Single source of truth (`ProfileMapper`) converting JPA Entities to safe DTO projections, preventing N+1 queries.

DESIGN PATTERNS APPLIED IN PRODUCTION

- Specification Pattern (Criteria API):** Dynamic multi-attribute predicate builder in `ProfileSpecification` for composite filtering (age, city, diet, gotra, marital status).
- Data Transfer Object (DTO) Pattern:** Decouples persistence models from external consumers, enforcing field-level validation and masking sensitive data.
- Mapper Pattern (ProfileMapper):** Centralizes entity-to-DTO conversion logic in a single component, eliminating duplicate mapping in `ProfileService` & `DiscoverService`.
- Template Method & Strategy Pattern:** Used in HTML email generation and dynamic membership tier entitlement computation based on active payment validity.

SOLID PRINCIPLES ADHERENCE

- Single Responsibility (SRP):** Separate services for Authentication, Profile, Discovery, Likes, Interests, Payments, and Stats.
- Open/Closed Principle (OCP):** CriteriaBuilder specifications allow adding new search predicates without altering core query engines.
- Liskov Substitution (LSP):** Generic repositories extend standard Spring Data `JpaRepository` and `JpaSpecificationExecutor`.
- Interface Segregation (ISP):** Focused API contracts for safe views, public stats, search result DTOs, and admin operations.
- Dependency Inversion (DIP):** High-level controllers and services depend entirely on abstractions injected via Spring IoC.

RELATIONAL DATABASE MODEL (MATRIMONIAL_DB)

Table Name	Key Attributes & Types	Relationships / Foreign Keys	Business Rules & Indexes
users	id (BIGINT PK), email (UK), password (BCrypt), role, enabled (BOOLEAN), created_at	1:1 with profiles, 1:N with payments, 1:N with otp_tokens	Email uniqueness index, soft-deletable with OTP verification cascade.
profiles	id (BIGINT PK), full_name, age, gender, marital_status, mobile_no, weight (DECIMAL 5,2), gotra, city, state	FK: user_id (UNIQUE). 1:N with profile_photos, 1:1 with expectations	Mobile masked unless viewed by owner or Level 3 VIP member.
profile_photos	id (BIGINT PK), photo_url, is_primary (BOOLEAN), created_at	FK: profile_id → profiles(id) ON DELETE CASCADE	Max 10 photos per profile, exactly 1 primary DP thumbnail enforced.
expectations	id (BIGINT PK), min_age, max_age, preferred_marital_status, preferred_gotra, preferred_diet	FK: profile_id (UNIQUE) → profiles(id)	Partner criteria with dynamic Add/Edit state and empty warning badge.
likes	id (BIGINT PK), sender_id, receiver_id, created_at	FK: sender_id, receiver_id → profiles(id)	Unique constraint (sender, receiver), daily like quota enforcement.
interests	id (BIGINT PK), sender_id, receiver_id, status (PENDING, ACCEPTED, DECLINED)	FK: sender_id, receiver_id → profiles(id)	Two-way interest requests, true mutual match evaluation (<code>MATCHED</code>).
payments	id (BIGINT PK), razorpay_order_id, razorpay_payment_id, amount, feature (VARCHAR 50), status, expires_at	FK: user_id → users(id) ON DELETE CASCADE	2-Month (60 days) validity timestamp cut-off checked on every feature call.

PRODUCTION HIKARICP DATABASE CONNECTION POOL SIZING

maximum-pool-size: 25

Calibrated for 5,000 registered users with 200 peak concurrent active sessions.

minimum-idle: 10

Pre-warmed idle connections preventing cold start latency on traffic spikes.

leak-detection-threshold: 2000ms

Proactively warns and flags slow queries or unclosed connections across JPA transactions.



1. REGISTRATION, TWO-STEP EMAIL OTP & SECURITY VERIFICATION

Designed for high conversion without compromising safety. Simplified signup requests only essential credentials initially, deferring detailed biographical questions to the Profile Setup stage.

- **Two-Step OTP Flow:** Form submission generates a cryptographically random 6-digit OTP token dispatched via JavaMailSender SMTP.
- **Mobile App Switch Preservation:** Pending registration state is cached in `localStorage` so switching to Gmail does not reload or wipe the OTP input screen.
- **Resend OTP Cooldown:** 60-second countdown timer throttles email spamming.
- **Inline Email Typo Correction:** "✍ Wrong Email? Change Email" button allows immediate corrections without restarting the registration journey.
- **Mandatory Consent & Legal Compliance:** Registration enforces explicit consent checkboxes linked directly to Privacy Policy and Terms & Conditions.
- **OTP-Protected Account Deletion:** Complete account purge requires dedicated email OTP verification with cascading cleanup.

2. PROFILE MANAGEMENT & CULTURAL COMMUNITY CUSTOMIZATION

6 Collapsible Accordions

Organized into Personal, Horoscope, Family, Contact, Lifestyle, and Expectations sections with intelligent auto-expand on form validation error.

Astrological & Family Nuances

3-button Mangal & Shani toggles (Yes / No / Prefer not to say), Gotra, 3-part birth time (HH:MM AM/PM), and mandatory Father & Mother names.

Precision Data & Privacy

2-decimal weight schema (`DECIMAL(5,2)`), masked contact numbers (`98*****`), and gated address protection.

3. 60 FPS INSTAGRAM-STYLE HIGH-PERFORMANCE PHOTO STUDIO

Engineered with HTML5 Canvas and native `requestAnimationFrame` sync, eliminating client-side lag and screen freezing across both desktop and mobile devices.

- **Multi-Photo Upload & Queue:** Upload up to 10 photos simultaneously with interactive thumbnail management and primary DP selector.
- **Multiple Aspect Ratios:** Supports 4:5 Portrait, 1:1 Square, Original, and 16:9 Cinematic Widescreen (badged with "★ Best View").
- **Interactive Transform Tools:** Smooth zoom slider, 90° clockwise rotation, and touch-optimized drag-to-pan repositioning.
- **Per-Photo State Persistence:** Save crop updates independently with green checkmark visual indicators and auto-disable safety buttons.
- **120Hz Display Synchronization:** Hardware-accelerated canvas exporting pixel-perfect cropped JPEG representations to the server.

4. SEARCH, MULTI-CRITERIA DISCOVERY & TRUE MUTUAL MATCHING

Dynamic JPA Search & Layouts

- **Multi-Parameter Filter:** Filter profiles by Age range (min/max), Marital status (Single, Divorced, Widowed), and Diet with live updates.
- **Grid vs. List View Toggle:** Users can switch between compact card grids and detailed vertical list cards, persisted in `localStorage`.
- **Live Community Bar:** Displays real-time statistics (total users, today's registrations) with an instant refresh trigger.

Two-Way Interests & True Mutual Matches

- **Two-Way Interaction Workflow:** Sending an interest allows the recipient to Accept, Decline, or Send Interest Back.
- **Mutual Match Recognition:** Status elevates to Mutual Match only when both profiles have accepted each other.
- **Withdraw Pending Interest:** Users can retract pending requests before action is taken, updating quotas dynamically.
- **Sent Interests Hub:** Dedicated dashboard categorizing sent requests into Pending, Accepted, and Not Interested.



3-LEVEL MEMBERSHIP MONETIZATION TIER SYSTEM (2-MONTH VALIDITY)

A sustainable community-funding model replacing ad bloat. Free accounts are capped at 10 profiles, incentivizing upgrades through feature-rich 60-day tiers with automated 18% GST calculation.

Tier / Plan	Pricing (60 Days)	Feature Entitlements	Visibility & Contact Access
Free Member	₹0 / Forever	10 Profile Discovery Cap, 1 Like/Day, 10 Interests total.	Standard ranking. Phone and address masked.
Level 1 — Essential	₹99 (Offer: ₹49)	Full profile browsing, 2 Likes/Day, 10 Interests, search filters.	Enhanced ranking. Masked contact details.
Level 2 — Premium	₹199 (Offer: ₹99)	Unlimited Likes & Unlimited Interest Requests, priority inbox.	High visibility ranking (Level 2 > Level 1 > Free).
Level 3 — VIP Crown	₹299 (Offer: ₹149)	All Level 2 privileges + Unlimited Biodata Studio PDF Downloads.	Top Visibility + Full Phone & Address Unmasking.

Dynamic Registration Offer: Users registering within 7 days receive special discounted promotional rates (₹49, ₹99, ₹149).

Bilingual Toggle: Entire membership page toggles seamlessly between English and Gujarati.

AUTOMATED MARRIAGE BIODATA STUDIO (8 AUTHENTIC THEMES)

A specialized client-side design engine generating luxury, print-ready 1-page A4 Marriage Biodatas auto-populated from user profile records and high-res cropped portraits.

1-Page Precision Fit (210mm x 297mm): Strictly calibrated font sizing and margins guaranteeing 100% single-page output with zero trailing overflow.

Instant Auto-Fill: Pulls Personal Bio, Horoscope/Kundali, Education, Career, Family, and Contact details instantly.

8 Handcrafted Traditional & Luxury Themes:

- Royal Heritage:** Traditional gold borders
- Vedic Shubh:** Sacred auspicious saffron
- Kesari Vintage:** Warm antique parchment
- Temple Elegance:** Divine marigold motif
- Vedic Gold:** Ornate festive gold scroll
- Royal Ruby:** Regal deep crimson velvet
- Royal Sapphire:** Majestic royal blue
- Classic Ivory:** Minimalist elegant ivory

ANTI-SCREENSHOT SAFEGUARDS & PAYMENT SECURITY

Anti-Screenshot Interception

Intersects keystrokes (`Win+Shift+S` , `PrtScn`) and triggers an immediate blur overlay to deter unauthorized captures of unpaid biodatas.

Focus-Loss Blur Shield

Monitors `window.onblur` and `visibilitychange` ; blurs biodata preview instantly whenever external screen recording tools are focused.

Cryptographic Payment Verify

Razorpay payments verified via server-side HMAC-SHA256 signature hashing before granting plan entitlements, preventing client tampering.



CORE PRODUCTION REST API DIRECTORY

Method	Endpoint Path	Auth Tier	Functional Responsibility
POST	/api/auth/register	Public	Generates pending user & sends 6-digit registration OTP.
POST	/api/auth/register/verify-otp	Public	Activates account, persists credentials, returns JWT bearer token.
POST	/api/auth/login	Public	Authenticates credentials via BCrypt; issues 24-hour JWT token.
GET	/api/profile/me	Bearer JWT	Retrieves current user's profile, expectations, and photo status.
PUT	/api/profile	Bearer JWT	Updates biographical, horoscope, family, and contact attributes.
GET	/api/profiles/{id}	Bearer JWT	Returns public profile safe view; mobile masked unless Level 3 VIP.
GET	/api/discover?minAge=&maxAge=&diet=...	Bearer JWT	Dynamic CriteriaBuilder multi-attribute search with pagination.
POST	/api/likes/{profileId}	Bearer JWT	Registers a like; enforces daily like quota based on active tier.
POST	/api/interests/{profileId}	Bearer JWT	Sends mutual interest request; checks active quota availability.
GET	/api/interests/sent?status=PENDING	Bearer JWT	Returns sent interest requests with status filter tabs.
POST	/api/payment/create-order	Bearer JWT	Generates Razorpay order with calculated dynamic pricing & GST.
POST	/api/payment/verify	Bearer JWT	Cryptographically verifies HMAC signature & unlocks membership.
DELETE	/api/account?otpCode=	Bearer JWT	OTP-protected permanent account and media purge with FK safety.

SECURITY & HARDENING STANDARDS

- SQL Injection Shield: 100% parameter binding through Spring Data JPA & CriteriaBuilder; zero raw SQL concatenation.
- CORS Policy & Host Binding: Strict origin filtering in production with server.host=true for local Wi-Fi mobile testing.
- Payload Sanitization: Jakarta Validation (@NotBlank , @Min , @Max) protects against buffer overflows and malformed inputs.
- Structured Logging: Logback XML configuration with daily rotation and RequestLoggingFilter tracking execution times.

PRODUCTION SCALING & FUTURE ROADMAP

- AWS S3 & CloudFront Integration: Transition from local disk photo storage to AWS S3 buckets with pre-signed CDN delivery.
- WebSocket Instant Messaging: Real-time 1-on-1 private messaging between mutual matches (MATCHED status).
- Admin Governance Console: Dedicated administrative dashboard for manual profile verification and fraud monitoring.
- Automated Kundali Matching: Algorithmic Guna Milan calculation based on planetary birth time and coordinates.

Pragnesh Maru

Java Spring Boot Developer | Full-Stack Software Engineer

Specializing in robust Spring Boot 3 enterprise backends, high-performance REST APIs, relational database design, clean architecture, and modern React SPAs.

Project: PrajapatiSamaj
Matrimonial

Location: Gujarat, India

Status: Ready for Production &
Deployment